

Yujie Zhang

Curriculum Vitae

Biography

- 2023 (Dec)- **Postdoctoral fellow**, *University of Waterloo, Institute for Quantum Computing/ Perimeter Institute for Theoretical Physics*, Waterloo, CA.
- 2017 -2023 **Ph. D. in Physics**, *University of Illinois at Urbana-Champaign, Department of Physics*, Urbana, USA.
- 2013-2017 **B. Sc. in Physics**, *Nanjing University, Kuang Yaming Honor School*, Nanjing, China.

Preprints

- 2025 **CritPt group**, *Probing the Critical Point (CritPt) of AI Reasoning: a Frontier Physics Research Benchmark*, arXiv:2509.26574 (2025)
- Y Zhang**, Y Ying, D Schmid, *Quantifiers and witnesses for the nonclassicality of measurements and of states*, arXiv:2504.02944 (2025) [Quantum accepted]

Publications

- 2026 **Y Zhang**, J Spodek, D Schmid, C Reid, T Jennewein, K, Resch, R Spekkens, *Entanglement certification using noncontextuality inequalities*, Phys. Rev. X (2026)
- Y Zhang**, D Schmid, Y Ying, R Spekkens, *Reassessing the boundary between classical and nonclassical for individual quantum processes*, Phys. Rev. X 16, 021050 (2026)
- S Vinet, M Clementi, M Bacchi, **Yujie Zhang** et al, *Time-resolved certification of frequency-bin entanglement over multi-mode channels*, npj Quantum Information 12, 38 (2026)
- 2025 **Y Zhang**, T Jennewein, *Criteria for optimal entanglement-enhanced long baseline imaging protocols*, Phys. Rev. Research 7, 043278 (2025)
- Y Zhang**, J Zhang, E Chitambar, *Cost of Simulating Entanglement in Steering Scenario*, Quantum 9, 1902 (2025)
- S Vinet, W Wu, **Y Zhang**, T Jennewein, *Feasibility study of frequency-encoded photonic qubits over a free-space channel*, Opt. Express 33, 40437-40449 (2025)
- Y Wang, **Y Zhang**, VO Lorenz, *Temporally Localized Quantum Operations on Continuous-Wave Thermal Light*, Phys. Rev. Lett. 135, 113602 (2025)
- Y Wang, **Y Zhang**, VO Lorenz, *Astronomical interferometry using continuous-variable quantum teleportation*, Phys. Rev. Research 7, 023154 (2025)

- K. Kapoor, S. Hoseini, J. Choi, B. E. Nussbaum, **Y Zhang** et.al, *Public Quantum Network: The First Node*, Appl. Phys. Lett. 126 (5): 054002 (2025)
- 2024 **Y Zhang**, E Chitambar, *Exact Steering Bound for Two-Qubit Werner State*, Phys. Rev. Lett. 132, 250201 (2024)
- X Chen*, **Y Zhang***, A Winter, VO Lorenz, E Chitambar, *Information Carried by a Single Particle in Quantum Multiple-Access Channels*, PRA 109, 062420 (2024)
- 2023 K Garay-Palmett, DB Kim, **Y Zhang**, F Domínguez-Serna, VO Lorenz, AB U'Ren, *Fiber-based photon-pair generation: tutorial*, JOSA B **40** (3), 469-490 (2023)
- D Diaz, **Y Zhang**, Y Wang, VO Lorenz, P Kwiat, *Toward Quantum-enhanced Interferometric telescoping*, Frontiers in Optics (2023)
- 2022 **Y Zhang**, X Chen, E Chitambar, *Building multiple access channels with a single particle*, Quantum **6**, 653 (2022)
- 2021 Y Wang, **Y Zhang**, VO Lorenz, *Superresolution in interferometric imaging of strong thermal sources*, Phys. Rev. A **104**, 022613 (2021)
- 2020 **Y Zhang***, RA Bravo*, VO Lorenz, E Chitambar, *Channel activation of CHSH nonlocality*, New Journal of Physics 22 043003 (2020)
- 2019 **Y Zhang**, R Spiniolas, K Shinbrough, B Fang, O Cohen, and VO. Lorenz, *Dual-pump approach to photon-pair generation: demonstration of enhanced characterization and engineering capabilities*, Opt. Express **27**, 19050-19061 (2019)
- 2018 A Zhang, **Y Zhang**, F Xu, L Li, and L Zhang, *Adaptive tomography of qubits: purity versus statistical fluctuations*, J.Phys.A: Math.Theor. **51** 395304 (2018)

Talks

- 2026 **YJ Zhang**, et al, "A paradigm for Entanglement certification using noncontextuality inequalities " Foxconn
- 2025 **YJ Zhang**, et al, "Criteria for optimal entanglement-enhanced long baseline imaging protocols" AQIS
- YJ Zhang**, et al, "Defining nonclassicality for individual quantum process" QPL
- 2024 **YJ Zhang**, et al, "Exact steering bound for two-Qubit Werner state," QIP
- 2023 **YJ Zhang**, et al, "Quantum enhanced interferometric imaging" Quantum 2.0
- 2022 XA Chen*, **YJ Zhang***, et al, "Information carried by a single particle in multiple-access channels" TQC
- 2021 **YJ Zhang**, et al, "Building multiple-access channel with one particle" Beyond IID
- 2020 **YJ Zhang**, et al, "Dual-Pump design enables novel photon-pair characterization and engineering" CLEO
- 2019 **YJ Zhang**, et al, "Tailored photon-pair generation in optical fiber through dual-pump spontaneous four-wave mixing" FIO-LS

* Co-first authors

Awards and Honors

2023 **Boeing Quantum Creators Prize.**

The Boeing Quantum Creators Prize ([Link](#)) recognizes early-career researchers for work that moves the field of quantum information science and engineering in new directions. The program began in 2021 and expanded in 2023 thanks to a commitment from Boeing. Prizes are awarded at the annual Boeing Quantum Creators Prize Symposium hosted during the annual Chicago Quantum Summit.

2019 **University Fellowship for outstanding research accomplishments.**

This graduate fellowship is awarded to a student who has demonstrated significant ability in research.

Research

Quantum foundations

2024–present **Generalized Contextuality**, *Robert Spekkens Group*, PI, Waterloo.

Developing a comprehensive framework for analyzing the non-classicality of individual quantum processes through operational contextuality in different causal structures and exploring its connections to other well-established notions of non-classicality in quantum theory:

- Introduced new definitions, certification methods, and quantification techniques for non-classical measurements and preparations, encompassing measurement incompatibility and steering assemblages as special cases.
- Developing a novel framework for defining the non-classicality of individual quantum processes in quantum circuits, namely different quantum networks, extending beyond standard notions such as entanglement and entanglement-breaking channels.
- Proposing new tools for certifying bipartite entanglement via operational contextuality and integrating these theoretical advancements with experimental implementations and data analysis. This aims to establish innovative methods for device-independent and theory-independent certification of the full range of entangled quantum states.

2018–2023 **Entanglement Theory**, *Eric Chitambar Group*, UIUC, Urbana.

Focused on the study of non-classical effects such as entanglement, steering, and nonlocality, with emphasis on their characterization, classification, and activation in quantum networks:

- Derived the exact quantum steering bound for qubit Werner states using positive operator-valued measures (POVMs), resolving an open question posed at IQOQI (Problem 39).
- Established rigorous lower bounds on the simulation cost for the class of unsteerable entangled states, demonstrating that (a) some unsteerable states require unbounded simulation costs, and (b) all entangled states incur a strictly greater simulation cost than separable states.
- Discovered a novel superactivation phenomenon in quantum nonlocality, showing that two noisy nonlocality-breaking quantum channels used in parallel can enable nonlocal quantum state distribution.

Quantum information theory

2023–present **Higher dimension quantum system**, *Thomas Jennewein Group*, UW, Waterloo.

We are working on an experiment-oriented theoretical approach for processing high-dimensional quantum systems encoded in frequency. We have developed a framework to understand and overcome the challenge of utilizing higher dimensional encoded frequency-bin state in large-scale quantum network, and are studying protocols for processing higher-dimensional quantum systems and their entanglement in practical quantum networks.

- 2023-present **Distributed quantum metrology**, *Thomas Jennewein Group*, UW, Waterloo.
We are studying and developing new protocols for distributed quantum metrology, using non-local resources in quantum networks to achieve better estimation of astronomical sensing, satellite positioning, and global time synchronization.
- Proposed a new framework for the study of shared quantum resources for quantum-enhanced telescoping. Provided several new quantum protocols for better astronomical sensing using novel shared entangled states, and linear optical quantum teleportation.
 - We aim to design theoretical frameworks to study quantum-enhanced metrology using complicated distributed linear optical interferometer and implement them in near-term quantum networks
- 2023-present **Quantum network theory**, *Eric Chitambar Group*, UIUC, Urbana.
At present, we are studying and developing new protocols for city-scale quantum networks, including genuine network nonlocality, network quantum steering and resource theory of linear optics. We aim to design near-term, experimentally friendly quantum applications for networks beyond the standard entanglement-swapping-based protocols.
- 2018–2022 **Quantum communication/coherence**, *Eric Chitambar Group*, UIUC, Urbana.
We conducted theoretical studies on quantum-enhanced communication in multiple-access channels (MACs), where the genuine multiple-level coherence of a single quantum particle was utilized as the resource:
- We proposed new communication protocols using MACs built with a single particle and discovered new quantum games to witness novel quantum advantages. This framework was applied to an N-port interferometer experiment in which interference beyond double-slit experiments was systematically discussed.
 - We analyzed the information communication rates of MACs built with a single particle and proposed quantum-enhanced communication protocols using a single photon and linear optical interferometry.
- Quantum optics**
- 2022–2023 **Quantum network experiment**, *Lorenz and Kwiat groups*, UIUC, Urbana.
Currently, we built the first publicly accessible fiber-based quantum network in collaboration with Kwiat's group for network quantum information processing in Urbana-Champaign. This involved the experimental development of an ultrabright, factorable, hyper-entangled photon-pair source using a bulk PPKTP crystal, and the characterization and compensation of noisy quantum links using underground optical fibers over tens of kilometers.
- 2019–2023 **Optical quantum sensing**, *Lorenz Group*, UIUC, Urbana.
We have designed and implemented quantum-enhanced interferometric imaging and discussed the emerging role of quantum technologies for improved astronomical observation:
- We analyze the decomposition of thermal light with a flat spectrum into localized pulses and rigorously examine the validity of this approximation for information processing in astronomical sensing applications.
 - We constructed a tabletop demonstrative experiment that integrates the original quantum telescope scheme proposed by Gottesman with techniques developed in classical optical telescoping, such as closure phase, leveraging these methods to enhance the performance of the quantum telescope.
 - Proposed a new quantum telescoping scheme utilizing continuous variable (CV) entanglement and CV teleportation, achieving improved performance in certain source intensity regions compared to the original quantum telescope scheme.
 - Introduced new super-resolution results on strong two-point thermal sources using an interferometric-based imaging scheme.
 - Explored the roles of reference frameworks in astronomical interferometric imaging, introducing a new protocol based on the generalization of Hanbury Brown-Twiss interferometry.

- 2017–2020 **Photonic state engineering**, *Lorenz Group*, UIUC, Urbana.
 We worked on engineering photonic quantum states in terms of their generation, control, and storage:
- Built a factorable single photon source using dual-pump spontaneous four-wave mixing (SFWM) and showed its enhanced characterization and engineering capabilities
 - Helped in building a hyper-entangled photon-pair source using SFWM in polarization maintaining fiber that supports few spatial modes and an ultra-broadband quantum memory with hot atomic barium.
- 2016–2017 **Quantum tomography/Talbot effect**, *Lijian Zhang Group*, NJU, Nanjing.
 I spent one year at Prof. Zhang's group studying optical quantum information processing:
- Analyzed adaptive quantum state tomography on the nearly pure quantum state.
 - Studied the Talbot effect with path-entangled states and found that its recurrence feature disappears with strong path entanglement.

Mentorship experience

Worked in close relation with several undergraduate and graduate students with different backgrounds at University of Illinois, University of Waterloo and Perimeter Institute:

- 2024- **Graduate student**, *Jonah Spodek*, at University of Waterloo, Insitute for quantum computing, Waterloo, CA.
 Project: Experiment test of entanglement with operational contextuality
- 2024- **Graduate student**, *Yile Ying*, at Perimeter Institute, Waterloo, CA.
 Project: Nonclassicality of general quantum process
- 2023- **Graduate student**, *Stéphane Vinet*, at University of Waterloo, Insitute for quantum computing, Waterloo, CA.
 Project: Frequency-bin encoded quantum system over free-space channel
- 2022-2023 **Undergraduate student**, *Soroush Hoseini*, at University of Illinois at Urbana-Champaign, Department of Physics, Urbana, USA.
 Project: Long-distance entanglement distribution with classical shared reference frame
- 2022-2023 **Graduate student**, *Keshav Kapoor*, at University of Illinois at Urbana-Champaign, Department of Physics, Urbana, USA.
 Project: Public quantum network: the first nodes
- 2021-2022 **Undergraduate student**, *Jiixin Zhang*, at University of Illinois at Urbana-Champaign, Department of Physics, Urbana, USA.
 Project: Simulation complexity of entanglement and compatible measurements
- 2019-2022 **Graduate student**, *Xinan Chen*, at University of Illinois at Urbana-Champaign, Department of Electrical and Computer Engineering , Urbana, USA.
 Project: Multiple-access communication with one single particle
- 2017-2019 **Undergraduate student**, *Rodrigo Araiza Bravo*, at University of Illinois at Urbana-Champaign, Department of Physics, Urbana, USA.
 Project: Super-activation of quantum non-locality in quantum network; Genuine multi-partite entanglement and nonlocality
- 2017-2018 **Undergraduate student**, *Ryan Spiniolas*, at University of Illinois at Urbana-Champaign, Department of Physics, Urbana, USA.
 Project: Photonic engineering of in-fiber single photon source

Outreach

2023 **Public Quantum Network at Urbana Free Library.**

Leading the creation of the Public Quantum Network (Link) that distributes entangled photons from UIUC to a local library, integrating research and education.

2023 **IQUIST at Ant-Man The Wasp: Quantumania.**

Led a post-screening public discussion on quantum science after Ant-Man and the Wasp: Quantumania; event coordinated by the Illinois Quantum Information Science and Technology Center (IQUIST).